

SHAHADDIN GAFAROV

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EDUCATION

University of Florida

Ph.D. in Computer Science; Advisor: Dr. Tamer Kahveci

2025 – 2028 (expected)

Gainesville, FL

University of South Florida

B.S. in Computer Science; Advisor: Dr. Yu Sun

2021 – 2024

Tampa, FL

EXPERIENCE

University of Florida

April 2025 – Present

Graduate Research and Teaching Assistant, Bioinformatics and Medical AI

Gainesville, FL

- Built the first benchmark that tests whether AI models predicting protein interactions can be fooled by small, deliberate changes to their data; showed such changes can push a leading model from 0.87 AUC to worse than random guessing.
- Designing GRAFT, a deep learning model built on a pretrained foundation model that predicts the genes a spatial transcriptomics experiment did not measure; most accurate of all published methods compared in early results.
- Led the peer-review revision of SPACT, a cancer survival model combining tissue images with genomic data: reran nearly all experiments and resolved every reviewer comment, leading to publication in *Medical Image Analysis*.
- Built the evaluation pipeline for PLATO, a method that predicts how gene networks change over time; it outperformed 4 leading methods on leukemia data (ACM BCB 2025).
- Lead Teaching Assistant for Advanced Data Structures (~75 students) and Enterprise Software Engineering Practices (~200).

University of South Florida

December 2023 – December 2024

Undergraduate Research Assistant, Robotics

Tampa, FL

- Designed 3 benchmarking protocols and 4 evaluation metrics for robots grasping several objects at once, tested on 3 robot hands and a human baseline (IEEE RA-L 2025).
- Developed deep learning models that estimate the 3D position and orientation of objects in a pile for robotic grasping.

MusicLessonHub

2022

Full-Stack Developer, early-stage startup

Boise, ID (Remote)

- Built front-end interfaces and back-end endpoints for a platform matching music students with teachers across the United States, shipping user-facing features alongside the design team.

AzeriMed LLC

2020 – 2021

Software Engineer, Databases

Baku, Azerbaijan

- Built and maintained the database layer for aptekonline.az, Azerbaijan's first online pharmacy (National Internet Award, 2023).
- Reduced query latency and error rates by redesigning database error handling and improving backend reliability.

PUBLICATIONS & MANUSCRIPTS

- Gafarov, S.**, Khatib, T., Yang, Z., Turan, M., Sun, J., Atay, Y., & Kahveci, T. *A Comprehensive Benchmark for Adversarial Robustness in Dynamic Biological Network Prediction*. Under review, ACM Computing Surveys.
- Khatib, T., **Gafarov, S.**, Baktir, N., Gallardo, A. M., Atay, Y., & Kahveci, T. *Biological Network Prediction: A Survey of Dynamic and Static Algorithms*. Under review, Archives of Computational Methods in Engineering (Springer).
- Öğülmüş, F. E., **Gafarov, S.**, Almaloğlu, Y., Özdemir, B. H., Ok Atılgan, A., Demir, D., Özen, Ö., Keleş, G. E., Kahveci, T., & Turan, M. (2026). *SPACT: A Clustering-Driven Multi-Modal Framework for Survival Prediction Using Genomic and Histopathology Data*. *Medical Image Analysis*, 112, 104078.
- Khatib, T., **Gafarov, S.**, Strikowski, J. D., Turan, M., & Kahveci, T. (2025). *PLATO: Predicting Longitudinally-Aligned Time Observations of Biological Networks*. *Proc. of the 16th ACM Int'l Conf. on Bioinformatics, Computational Biology and Health Informatics (BCB '25)*.
- Chen, T., Frumento, R., Pagnanelli, G., Cei, G., Keth, V., **Gafarov, S.**, Gong, J., Ye, Z., Baracca, M., D'Avella, S., Bianchi, M., & Sun, Y. (2025). *Benchmarking Multi-Object Grasping*. *IEEE Robotics and Automation Letters*, 10, 9510–9517.

LANGUAGES & TECHNOLOGIES

Languages	Python, C/C++, Java, SQL, JavaScript
Machine Learning	PyTorch, scikit-learn, NumPy, Pandas, Matplotlib, Scanpy
Infrastructure	Slurm, HPC clusters, Git, Docker, CUDA

AWARDS

Full-Ride Scholarship	University of Florida
Green & Gold Presidential Award	University of South Florida
Bronze Medal	National Informatics Olympiad of Azerbaijan